DORTMUND OFF-GRID AIR — Part 1: Website Slides (#1 to #11)

Exact Spoken Script Matching Every Slide on `soulnassim.github.io/Dortmund-OFF-grid-air/pitch.html`

SLIDES #1 – #11

Slide #1 · Title — `Dortmund OFF-Grid Air`

0:00 – 0:25 (25s)

🎬 **ON SCREEN: pitch.html#1 (3D Isometric Dortmund Hero Animation). Pause 2s, smile, look at the jury.**

"Imagine ten autonomous drones flying over Dortmund right now — delivering urgent blood supplies between hospitals, or helping firefighters see a big city fire from above.

Normally, those drones rely on public 5G or the internet to stay connected. ***But what happens during a real crisis when cell towers go down, or an attacker targets the network?*** *(1s pause)*

That is exactly where our mission, Dortmund OFF-Grid Air, comes in: we have to keep all ten drones flying and connected back to HQ at the Reinoldikirche with **zero public internet and zero cell towers** — which means they have to build their own private network right in the sky!"

Slide #2 · The Problem — `Today's fleets rely on 2.4 GHz, 4G/5G & GPS`

0:25 – 0:50 (25s)

🎬 **ON SCREEN: pitch.html#2. Press [→] 3 times to reveal the 3 columns (`01`, `02`, `03`).**

"Why do commercial drones fail immediately in a crisis today? Because three things break at once:

• **First (`01 · PUBLIC SPECTRUM — 2.4 GHz & 4G/5G jam`):** Standard drones fly on public 2.4 GHz radio or public 4G/5G cell towers. A cheap radio jammer or power blackout wipes both out in seconds.

• **Second (`02 · STAR TOPOLOGY — No relay, no QoS`):** Every drone only talks straight to the ground and **cannot bounce signals through neighbor drones (no relay)** — so one tall building cuts it off! Even worse, there is **no traffic priority (no QoS)**, meaning heavy HD camera video clogs the weak link and chokes out tiny steering commands.

• **And third (`03 · BLIND NAVIGATION — GPS spoofed, alarms late`):** Standard autopilots blindly trust fake GPS coordinates, while standard rolling-average alarms stay blind to slow-ramping attacks for over **9 seconds**!"

Slide #3 · Our Answer — `Optical lasers beat the jammer. Now the city is the obstacle.`

0:50 – 1:15 (25s)

🎬 **ON SCREEN: pitch.html#3. Press [→] 3 times to reveal `22–28 m`, `120 m`, and `3 s`.**

"**To defeat radio jammers completely, we move our primary connection onto invisible 1550 nm Free-Space Optical (FSOC) lasers!** Because a laser is a narrow beam of light — not a radio wave — a radio jammer on the ground cannot jam it!

However, lasers create a new challenge: Dortmund's 3D buildings. While typical Dortmund blocks are **22 to 28 meters** tall, landmarks like the **Dortmunder U-Tower reach 95 meters**. When `DR-03` flies at 50 meters behind the U-Tower, the building blocks the laser beam!

Therefore, we place **two High-Altitude Relay Drones (`DR-01` and `DR-02`) at 120 meters** — high above every roof in Dortmund — with a **3-second hysteresis** so links never flicker on building edges."

Slide #4 · Hybrid Mesh — `Two off-grid links per drone (FSOC + DECT NR+)` 1:15 – 1:45 (30s)

🎬 **ON SCREEN: pitch.html#4 (Side-by-side FSOC 1550 nm vs DECT NR+ 1.9 GHz + 225 g Pod Bar).**

"Every drone carries a lightweight **225-gram hybrid pod** (signed with HMAC-SHA256) running **two off-grid channels that protect each other**:

• **First, when the sky is clear**, drones connect over **green FSOC Lasers (1550 nm)** at **1 Gigabit per second** and 2 milliseconds latency — streaming live HD video, steering, and telemetry with **0% radio jamming loss**.

• ***But what happens when a building or thick fog blocks the laser (>520m Mie scattering)?*** *(1s pause)* Automatically, it switches to **orange DECT NR+ (1.9 GHz)** — **the world's first license-free 5G-standard radio that lets drones build their own private network in the sky (with zero SIM cards, zero monthly fees, and zero cell towers)**, bending around buildings, penetrating thick fog 100%, and bouncing steering commands and 3D positions from drone to drone up to our 120-meter relays!

• ***And what if the opposite happens — what if an attacker jams the radio?*** Then any drone with clear Line-of-Sight stays 100% immune on the green laser!"

Slide #5 · Layer 1: Jamming Physics — `Distance decides. We don't cheat it.` 1:45 – 2:10 (25s)

🎬 **ON SCREEN: pitch.html#5. Press [→] 3 times to reveal the 3 rows (Green, Orange, Purple).**

"When an attacker turns on a jammer near the U-Tower, jamming power drops with distance ($1/r^2$), creating three regimes:

1. **With clear Line-of-Sight (Green):** The laser holds with **0% loss** because photons ignore radio noise.

2. **Behind a building in the Outer Zone (`>150 m`, Orange — `DR-07`):** The building blocks the laser, so DR-07 switches to **DECT NR+ backup radio**. Because radio has less bandwidth than a laser, our **QoS Emergency mode** automatically cuts off heavy video (**`Video 0%`**) so **C2 steering commands stay locked at 100% (12 kbps)**!

3. **And behind a building inside the Jammer Core (`<150 m`, Purple — `DR-03`):** Here, the U-Tower blocks the laser **AND** the jammer crushes the radio at the same time. Both links fail honestly."

Slide #6 · Autonomous Recovery — `When every radio fails, climb.` 2:10 – 2:35 (25s)

🎬 **ON SCREEN: pitch.html#6. Press [→] 4 times to reveal `0.0 s`, `1.5 s`, `>95 m`, and `3.2 s`.**

"***So what does DR-03 do when both the laser and radio are blocked at `0.0 s`?*** *(1s pause)*

After **1.5 seconds** isolated, DR-03 switches to its **onboard internal movement and camera sensors (`INS/VIO`)** — needing zero GPS and zero radio — and **automatically climbs straight up** from 50 meters to 125 meters!

The moment it clears the **95-meter U-Tower roof**, it catches clear Line-of-Sight to our **120-meter relay drone (`DR-01`)**, and in **3.2 seconds**, the green laser snaps right back online!"

DORTMUND OFF-GRID AIR — Slides #7 to #11 + Live Demo Walkthrough

SLIDES #7 – #11 + DEMO

Layer 2 (Contrastive AI) · Layer 3 (LLM Co-Pilot & Collaborative Learning) · Results · Team · Live Demo

Slide #7 · Layer 2: Threat Detection — `Catch the slow attack at onset (9.5s vs 0.0s)`

2:35 – 3:10 (35s)

🎬 **ON SCREEN: pitch.html#7. Press [→] to reveal `9.5s vs 0.0s` and the 3 bottom boxes.**

"Physics keeps our drones flying — ***but how will tomorrow's smart attackers try to bypass standard threat detectors?*** *(1s pause)*

Instead of turning a jammer on at full power all at once, a smart attacker **increases the jamming power very slowly over time (+1.55 dB/s) — a Ramping Jammer**. ***Why does that fool a standard threshold detector?*** Because a standard detector only compares the current signal to the **moving average of the last 4 seconds**. When the jamming increases slowly, that 4-second average shifts down right along with the attack — just like your eyes adjusting to a slowly darkening room — leaving the detector completely blind for **9.5 seconds**!

Our Contrastive Prototype Detector fixes this: at takeoff, it freezes a **healthy baseline** of 4 values (Signal Strength, Signal Clarity, Packet Loss, and GPS-to-INS offset). Because that takeoff baseline never shifts, we catch even a slow ramping jammer or slow GPS drift at **0.0 seconds** with **zero false alarms**!

And from my security research discussions at **MPI-SP**, **real trust comes when AI is honest about what it does NOT know**: when an attacker combines Jamming and GPS Spoofing in a brand-new pattern, our detector refuses to guess and flags **UNEXPLAINED**."

Slide #8 · Layer 3: LLM Co-Pilot — `The LLM never detects. It explains.`

3:10 – 3:45 (35s)

🎬 **ON SCREEN: pitch.html#8. Press [→] across the 4 pipeline nodes (`4D telemetry` → `Detector` → `Verdict` → `LLM analyst`).**

"**And because an `UNEXPLAINED` attack needs a label, that is where our LLM Security Co-Pilot steps in!**

Look at our four-step pipeline: **we never let an LLM fly the drone or do the detection** — detection is fast, deterministic math on our 4D telemetry. The LLM sits strictly downstream **offline at HQ** (with live university LLM support via one environment variable): it reads the detector's verdict, writes a **two-sentence German briefing**, opens a **prioritized `CRIT` ticket**, and gives the commander a **one-click `Confirm Threat` button**.

Watch what happens when the commander clicks `Confirm Threat` once: it triggers **Collaborative Fleet Learning**! HQ takes that tiny **4-number threat fingerprint** from the affected drones and **broadcasts it to all 10 drones in the fleet** — so every other drone immediately recognizes that attack at 0.0 seconds and proactively switches to its DECT backup route and INS lock before entering the danger zone (keeping recovery under 3 seconds) — **drastically cutting fleet downtime, lowering operational costs, and keeping hospital deliveries on schedule for clients like Morpheus Logistik!**"

Slide #9 · Results & F&E Roadmap — `What it adds up to.`

3:45 – 4:00 (15s)

🎬 **ON SCREEN: pitch.html#9. Press [→] to reveal the 4 KPIs (`10/10`, `3.2s`, `0.0s`, `225g`) and the 3-Phase Roadmap.**

"In numbers: **10 out of 10 drones stay reachable** under jamming, node loss, and dense fog; **3.2-second worst-case recovery** via autonomous Pop-Up; **0.0-second slow-attack detection**; and a **225-gram hardware-agnostic pod** that plugs into Morpheus Leitstand via MAVLink 2 — with a clear 12-month roadmap from COTS DECT NR+ bench testing to a 10-drone BVLOS trial between Klinikum Mitte and Westfalenhallen."

Slide #10 · The Team — `Built by engineers who ship under fire.`

4:00 – 4:10 (10s)

🎬 **ON SCREEN: pitch.html#10. Press [→] 3 times to show Nassim, Ilias Amir, and Sami.**

"Built in 24 hours by **Nassim Soultana** (3D Aerospace, FSOC/DECT mesh, Fog Physics & INS/VIO Pop-Up), **Ilias Amir Soultana** (Contrastive Prototype AI, HMAC-SHA256 Zero-Trust & LLM Co-Pilot), and **Sami Soultana** (4-tier QoS Video Shedding, 225g SWaP & Morpheus Leitstand Operations)."

Slide #11 · Live Demo — `Let's break it. Live.` → Left-to-Right Walkthrough on `index.html`

4:10 – 5:00 (50s)

🎬 **ACTION: On Slide #11, click the green button [Open mission control → index.html] and walk Left-to-Right!**

"Now let's click **Open Mission Control** and break it live, going from left to right!

• **1) Left Sidebar:** Here are all 10 drones (`10.41.0.1` to `10`) talking to HQ over green **FSOC Lasers** and orange **DECT NR+**.

• **2) Center 3D Map (Press `[J] Jamming` + `[W] Dense Fog`):** Look at the three rings around the U-Tower — outside drones stay green; in the outer ring, `DR-07` switches to DECT NR+; in the yellow ring, drones lock onto **INS** to reject fake GPS; and in the inner purple core behind the U-Tower, `DR-03` automatically **climbs from 50m to 125m (`POP-UP`)** to restore its laser in **3.2 seconds**!

• **3) Right Top & Middle (Resilience & QoS):** Top 4 boxes show **100% Reachability** and **3.2s Recovery**. Below, under **QoS Traffic Priorities** (`C2 Command`, `Telemetry`, `Inter-Drone` 3D anti-collision/laser aiming, and `Video`), heavy video is cut to **`0% (SHED)`** so **C2 Steering stays locked at 100%**!

• **4) Right Bottom (Click `[🎯 Coordinated RF+GNSS]` → Click `[✓ Confirm Threat]`):** `DR-03` and `DR-07` flag yellow **UNEXPLAINED**, our **LLM Co-Pilot** explains it in German and files a `CRIT` ticket, and with **one click on `Confirm Threat`**, **Collaborative Fleet Learning** broadcasts the 4-number fingerprint from HQ so all 10 drones turn **blue**!

The future of urban logistics is airborne — let's make sure it never falls out of the sky. Thank you!"

Technical Deep-Dive Notes (For Jury Q&A Only)

How Z-Score, Cosine Similarity, Detector-to-LLM JSON, and Hardware Scope work under the hood

Q&A READY

1. How Z-Score (>= 4.5) and Cosine Similarity (>= 0.84) Work Under the Hood

• **Step 1 — Anomaly Detection (Z-Score):** During the 40s (80-tick) takeoff warmup, each drone records the mean (μ) and standard deviation (σ) of 4 sensors: `[RSSI, SNR, Packet_Loss, GNSS_INS_Offset]`. Every 0.5s tick, we compute the 4D Z-score deviation vector `[z_rssi, z_snr, z_loss, z_gnss]`. If its Euclidean length `z_score >= 4.5` (e.g., `z = 24.6`), an attack is detected at `0.0s`.

• **Step 2 — Attribution (Cosine Similarity):** We compare the *direction* of that 4D Z-vector against known attack prototypes using Cosine Similarity (`sim` from `0.00` to `1.00`). Pure `jamming` points along `[High, High, Med, 0]`; pure `spoofing` points along `[0, 0, 0, High]`. When both hit together (`Coordinated RF+GNSS`), the vector points diagonally between them: max similarity is only `sim = 0.68 < 0.84` threshold → flagged as `UNEXPLAINED`.

2. How the Detector Feeds the LLM & How 1-Click Collaborative Learning Works

• **Detector → LLM JSON:** The detector passes a structured JSON payload `{device: "DR-07", label: "UNEXPLAINED", z_score: 24.6, dims: {rssi: 8.1, snr: 18.2, loss: 4.0, gnss_offset_m: 16.4}, similarity: 0.68}` to the LLM Analyst (`llm_analyst.py`), which runs offline at HQ (with optional live API mode to `gpt-6-studierende` on `chat.kiconnect.nrw/v1/responses`).

• **1-Click Fleet Learning (`CONFIRM_THREAT`):** Clicking Confirm Threat normalizes the 4D Z-vector into a unit vector (4 floats = 16 bytes) and saves it in the shared fleet prototype table as `coordinated_rf_gnss`. On the very next tick, similarity jumps from `0.68` to `0.99 >= 0.84` across all 10 drones.

3. Why Only 4 Sensors in the Detector? & How Do Moving Drones Aim Lasers?

• **Why we excluded Latency & Velocity from the 4D detector (Invariance):** When a drone flies behind the U-Tower and switches from FSOC Laser to DECT radio, its latency jumps from 5ms to 40ms. If latency or speed were inside the cyberattack detector, every building in Dortmund would trigger a false cyberattack alarm! Keeping only link-budget & navigation residuals gives us **0 false alarms** across 1,200 noisy drone-ticks.

• **How drones aim FSOC lasers (PAT) & Hardware Economics:** Drones broadcast their 3D GPS/INS coordinates over omnidirectional **DECT NR+** (`Inter-Drone` channel) so gimbals know where to aim the laser. And because our MEMS optical gimbal weighs 155g and our Nordic nRF9151 DECT NR+ radio weighs 70g (total **225g / 9.7W** hybrid pod), our architecture puts high-altitude relays on `DR-01` and `DR-02` at 120m while allowing lower drones to use DECT NR+ seamlessly.